Pneumonia Detection Using Chest X Ray Images

CAP 5516 Medical Image Computing | Assignment 1

1. Implementation Details

Both tasks use ResNet 18, a lightweight convolutional neural network with residual connections, as the backbone. The final fully connected layer is replaced with a 2 class output head (NORMAL vs. PNEUMONIA). In Task 1.1, all weights are randomly initialized and the network learns entirely from the chest X ray training data. In Task 1.2, the network is initialized with weights pre trained on ImageNet, then fine tuned on the same training data. This allows the model to leverage general visual features already learned from millions of natural images.

Data augmentation was applied during training to improve generalization: random horizontal flips, small rotations, and color jitter. All images were resized to 224 x 224 and normalized using the ImageNet mean and standard deviation.

Parameter	Value
Architecture	ResNet 18
Input size	224 x 224
Batch size	32
Epochs	20
Optimizer	Adam (lr = 1e 3, weight decay = 1e 4)
LR Scheduler	StepLR (step size = 7, gamma = 0.1)
Loss	Cross Entropy
Augmentation	Random flip, rotation +/ 10 degrees, color jitter

2. Training and Validation Loss Curves

Figure 1. Training and validation loss curves for Task 1.1 (left) and Task 1.2 (right).

Figure 2. Training and validation accuracy curves for Task 1.1 (left) and Task 1.2 (right).

Both models show a consistent downward trend in training loss across 20 epochs, indicating successful learning. The fine tuned model (Task 1.2) starts with a noticeably lower initial loss and reaches convergence faster, which reflects the advantage of starting from meaningful weight values rather than random noise. Training accuracy for both models climbs steadily and plateaus above 98% by the final epochs.

The validation curves appear erratic in both tasks. This is not a sign of poor training but rather a consequence of the extremely small validation set, which contains only 16 images. A single misclassification shifts validation accuracy by over 6%, making the curve noisy and unreliable as a standalone metric. Despite this, the best checkpoint strategy (saving the model with the highest validation accuracy seen so far) still produces reasonable test set performance.

3. Classification Accuracy on the Test Set

Task	Description	Best Val Acc	Test Accuracy
1.1	ResNet 18 from scratch	93.75%	80.61%
1.2	ResNet 18 fine tuned (ImageNet)	100.00%	82.37%

Fine tuning outperforms training from scratch by approximately 1.8% on the test set. While this margin may seem modest, it is consistent with the expectation that pre trained features provide a useful starting point even for a domain as different from ImageNet as medical imaging. Both models were trained for the same number of epochs under identical conditions, making the comparison fair.

Per class results for Task 1.1 (from scratch):

Class	Precision	Recall	F1 Score	Support
NORMAL	0.99	0.49	0.65	234
PNEUMONIA	0.76	1.00	0.87	390
Weighted avg	0.85	0.81	0.79	624

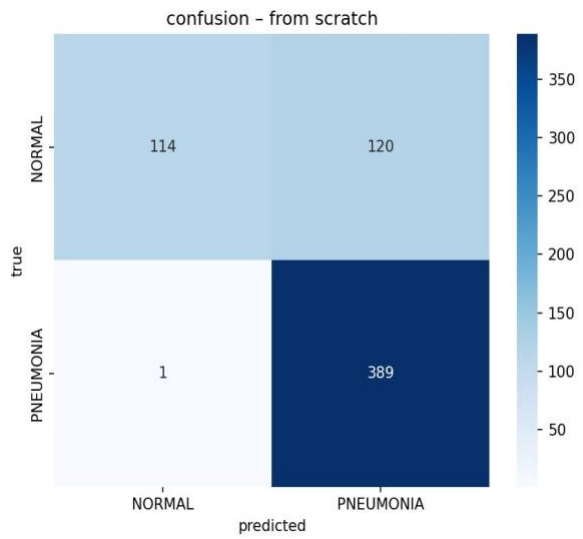

Figure 3. Confusion matrix for Task 1.1 (from scratch).

Per class results for Task 1.2 (fine tuned):

Class	Precision	Recall	F1 Score	Support
NORMAL	0.98	0.54	0.70	234
PNEUMONIA	0.78	0.99	0.88	390
Weighted avg	0.86	0.82	0.81	624

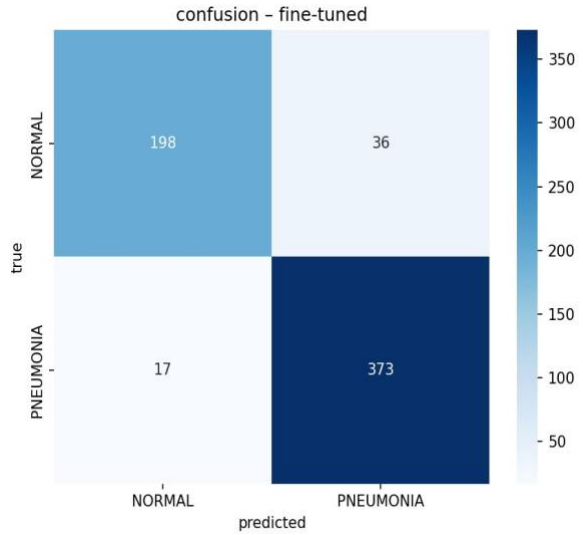

Figure 4. Confusion matrix for Task 1.2 (fine tuned).

A key observation across both models is the large gap between PNEUMONIA recall (99 to 100%) and NORMAL recall (49 to 54%). Both models are strongly biased toward predicting pneumonia. This is

partly explained by the class imbalance in the training set, where pneumonia cases outnumber normal cases roughly 3 to 1. The model learns that predicting pneumonia is the safer default. In a clinical screening setting this behavior is actually preferable since missing a pneumonia case (a false negative) carries far greater risk than a false alarm that prompts further investigation. That said, the low NORMAL recall does indicate that the model would benefit from class balancing strategies such as weighted loss or oversampling.

Fine tuning reduces the number of false positives on normal cases from 120 to 36, nearly a four fold improvement, while maintaining near perfect pneumonia recall. This is a meaningful clinical gain.

4. Failure Cases

All observed failure cases are false positives: normal X rays incorrectly classified as pneumonia. No false negatives (missed pneumonia cases) appeared among the top misclassified examples for either model. Examining the images reveals why these cases are difficult. Many of the misclassified normal X rays show increased opacification, slight haziness, or asymmetric lung density that visually resembles early stage or mild pneumonia. These subtle patterns can be ambiguous even for trained radiologists.

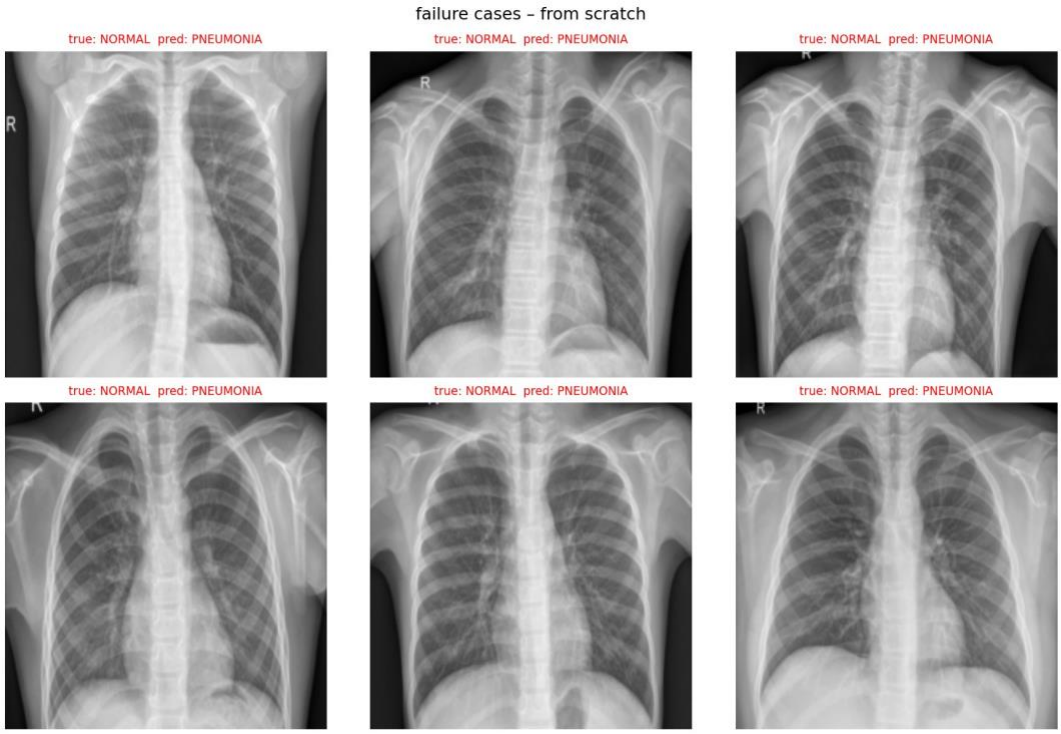

Figure 5. Failure cases for Task 1.1 (from scratch). All shown are NORMAL X rays misclassified as PNEUMONIA.

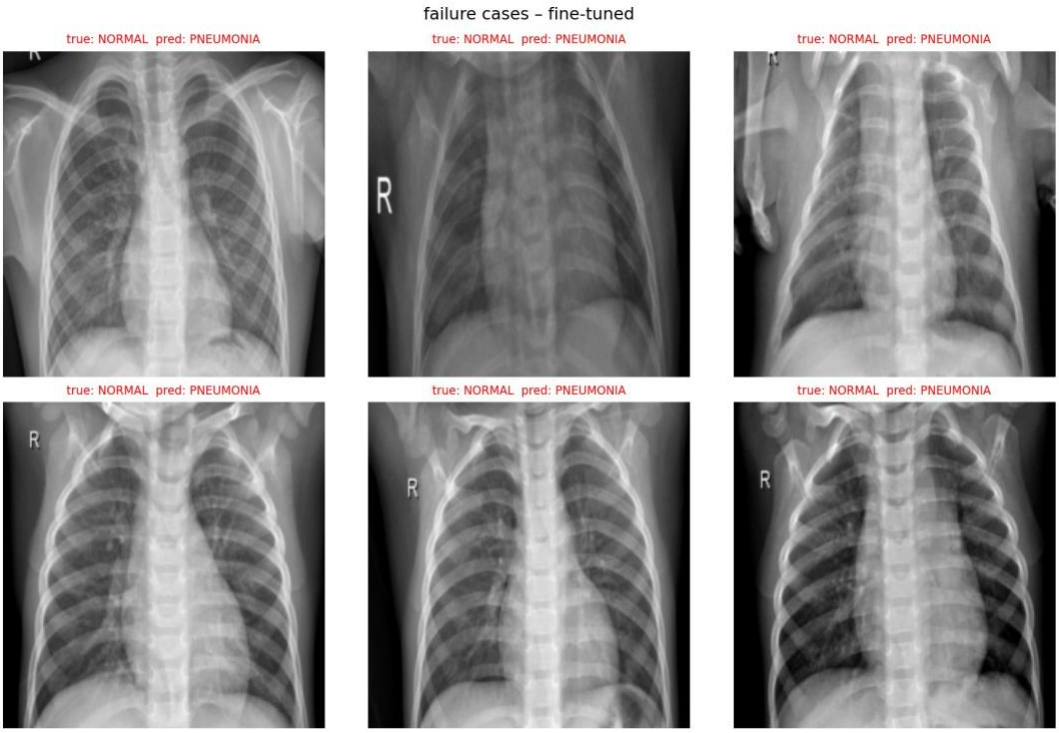

Figure 6. Failure cases for Task 1.2 (fine tuned). All shown are NORMAL X rays misclassified as PNEUMONIA.

Comparing the two sets of failure cases, the fine tuned model produces fewer total misclassifications on normal cases (36 vs. 120), and the misclassified images in Figure 6 tend to be visually more ambiguous than those in Figure 5. This suggests that fine tuning not only improves overall accuracy but also makes the remaining errors more understandable and harder to avoid without additional clinical context.

5. Conclusion

This assignment demonstrated the application of ResNet 18 to binary pneumonia classification from chest X ray images under two training regimes. Fine tuning a pre trained model consistently outperformed training from scratch, achieving 82.4% test accuracy compared to 80.6%, with a substantial reduction in false positives for normal cases. Both models showed near perfect sensitivity for detecting pneumonia, which is the clinically critical metric in a screening context.

The primary limitation is the heavy class imbalance in the dataset, which drives both models to over predict pneumonia at the cost of NORMAL recall. Addressing this through weighted cross entropy loss or data resampling would likely improve balanced performance. Additionally, the 16 image validation set is too small to reliably guide model selection, and a larger held out set would make the training process more stable and interpretable.

Overall, the results confirm that transfer learning is an effective strategy for medical image classification tasks, even when the source domain (ImageNet) differs substantially from the target domain (chest X rays).

Appendix: AI Use Disclosure

AI Tool	How It Was Used
Claude (Anthropic)	Helped me write and organize the training script (pneumonia_detection.py), covering model creation, the training loop, evaluation steps, and plotting. Also provided guidance on setting up and running the project on the Newton HPC cluster.

References

https://github.com/obeecodes/pneumonia-detection-resnet18